

Adapting Transformers for Light Dark Matter Event Reconstruction

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Using Artificial Intelligence for Detecting Dark Matter

Dark matter makes up ~85% of the matter in the universe and yet has not been directly detected in a lab. The existence of dark matter is supported by an overwhelming amount of evidence such as stars orbiting around the outer region of the Milky Way much faster than all the visible matter would suggest. The standard model of particle physics cannot explain gaps like this so if we can detect dark matter, it would revolutionise what we know about the universe.

So, if it's seemingly everywhere and extremely important then how come we have not detected it directly yet? The lack of detection is certainly not for a lack of trying and physicists have been developing increasingly sophisticated detectors. These detectors are often set up with light, 'target', materials such as ultra cold helium which maximises the amount of kinetic energy transferred when an incoming particle collides with the target. The more energy transferred by the incoming particle, the larger the signal and the easier it should be to detect a possible dark matter particle.

The huge challenge with modern dark matter direct detection experiments is that physicists have already ruled out most of the heavier, easier to detect, particles that might have been dark matter. This means that what's left are extremely light and weakly interacting particles, often ~10 times lighter than a proton. Particles like this are just what the upcoming experiment, DELight, will be looking for.

DELight will use 56 quantum sensors, 37 above the surface of the helium and 19 below. Each of these sensors records subtle temperature changes over time in which extremely small signals are buried in noise. But if these signals are so small we need quantum sensors just to record them, how can we get any information out of them? How can we tell where one of these collisions took place, how much energy was transferred or the kind of particle colliding?

Getting answers to these questions through a process called event reconstruction is where artificial intelligence (AI) comes in. By taking advantage of a type of neural network model called a Transformer we can extract these almost impossibly small signals from detector readings. This kind of model forms the basis of almost all large language models (LLMs) such as ChatGPT and Claude Opus. By harnessing the same algorithms used to capture the structure of natural language we can instead teach the transformer model to learn the physical processes that come after a collision.

We trained the transformer model to perform event reconstruction on hundreds of thousands of simulation-generated events. The result is that our adapted transformer achieves a lower energy resolution, more precise prediction of event locations and an improved ability to classify the collisions over the baseline model.

By hiding different sensor channels from the model and looking at the results, we have also shown that it's not just getting lucky picking up patterns but also genuinely learning the underlying dynamics. Some sensors hurt the performance and others less, just as predicted by the physics.

We aren't limited to just transformers. By combining a mixture of classical statistics and modern AI we can reconstruct dark matter events down to deposited energies as low as 10 eV, around the same as a handful of photons hitting your skin. This pushes the boundary of what is currently possible in reconstructing dark matter events, meaning we are one step closer to determining the nature of dark matter once and for all.

Abstract

The upcoming direct detection experiment, DELight, will use superfluid ^4He as a target material to search for light dark matter (LDM), at sub-GeV masses. Extracting recoil event information from sensor traces, event reconstruction, is critical for placing constraints on LDM candidates. Baseline approaches apply optimum filters (OF) to each trace to estimate signal features. These values are passed to a boosted decision tree (BDT). Predicting the position, recoil energy E_R and recoil type, i.e. electronic recoil vs nuclear recoil, at E_R from 10 eV to 500 eV requires encoding non-linear spatiotemporal correlations between sensor channel readings. We adapt and train transformer models on simulated DELight trace data to capture the non-linear correlations required for event reconstruction. We find that by injecting OF features into the transformer we achieve lower position RMSE, better energy resolution and higher ROC-AUC than the BDT baseline across all evaluated E_R . We additionally conduct a sensor channel masking analysis to better interpret the transformer’s capacity to learn the underlying signal partitioning physics.

1. Introduction

1.1. Dark Matter and Direct Detection

Dark matter (DM) is a non-luminous form of matter which interacts primarily through gravity and whose existence is strongly supported by cosmological observations of galaxy rotation curves [1] and gravitational lensing [2]. DM’s abundance, constituting 85% of all matter in the universe, is well constrained by parametric fits to cosmic microwave background observations [3]. Despite this abundance, DM has not been detected directly.

Thermal candidates are DM candidates which were in thermal equilibrium with the early universe plasma. Thermal candidates with a larger annihilation cross section, $\langle\sigma v\rangle$, will have a correspondingly smaller relic abundance, $\Omega_\chi h^2$, the density of DM surviving from the early universe [4]. A subset of thermal candidates with masses and interaction strengths characteristic of the weak nuclear interaction, weakly interacting massive particles (WIMPs) [5], have a predicted $\langle\sigma v\rangle$ such that the corresponding $\Omega_\chi h^2$ matches the Planck value $\Omega_\chi h^2 \approx 0.12$ [3]. This coincidence, known as the WIMP miracle, has motivated an extensive search for WIMPs through direct detection experiments. Direct detection experiments aim to record sparse scattering events of incident DM particles off target nuclei, and their results have led to the rejection of most viable WIMP models [6].

Advancing beyond WIMPs into probing for sub-GeV DM, light dark matter (LDM), comes with additional challenges. LDM kinematics result in energy depositions, E_R , when recoiling off target nuclei, orders of magnitude smaller than their WIMP counterparts, motivating the development of novel detector technologies.

The use of superfluid ^4He as a target material is a development which provides notable benefits to LDM direct detection. Superfluid ^4He ’s low nuclear mass relative to typical direct detection targets, such as xenon and silicon, provides better kinematic matching, facilitating sensitivity to LDM. Additionally, energy depositions from particles incident on the superfluid ^4He targets are partitioned into three independent signal channels. These three signal types, triplet signals from long-lived

triplet helium dimers, ultraviolet (UV) scintillations and quasiparticles [7] are critical for event reconstruction as is discussed in Section 1.2. These signal channels all have different characteristic arrival timescales. UV photons are emitted promptly following the decay of a singlet helium dimer, arriving at a timescale of $\mathcal{O}(\text{ns})$ [8]. Triplet signals are emitted with the decay of long-lived triplet helium dimers, arriving after $\mathcal{O}(1 - 10 \text{ ms})$ [9]. Finally, quasiparticle signals propagate ballistically before reaching the surface of the superfluid ^4He with an overall arrival timescale of $\mathcal{O}(\text{ms})$ [10].

The direct detection experiment, DELight [11], which this project supports, uses superfluid ^4He as a target material. The HeRALD [12] experiment similarly employs superfluid ^4He targets but differs in that it is instrumented with transition edge sensors (TESs) [13] in contrast to DELight’s magnetic microcalorimeters (MMCs) [14]. The DELight detector employs 56 of these MMCs with 37 positioned above the surface of the superfluid ^4He , top sensors, and 19 submerged beneath it, bottom sensors [10].

Quasiparticles produced following a recoil event are not directly detectable within the superfluid ^4He due to the high thermal resistance at the interface between the superfluid ^4He and the solid MMC sensors [15, 16]. Quasiparticle signatures are instead recorded only by top sensors through depositions from helium atoms [10] liberated from the superfluid ^4He surface through quantum evaporation [17, 18].

The energy threshold, E_{min} , is the minimum detectable E_R above the background noise floor. A lower energy threshold permits sensitivity to lower DM masses. DELight will probe for LDM with masses, m_χ , below 100 MeV and an anticipated energy threshold, E_{min} , of 20 eV for the first phase and 10 eV for the second phase of the experiment [11] (see Figure 1).

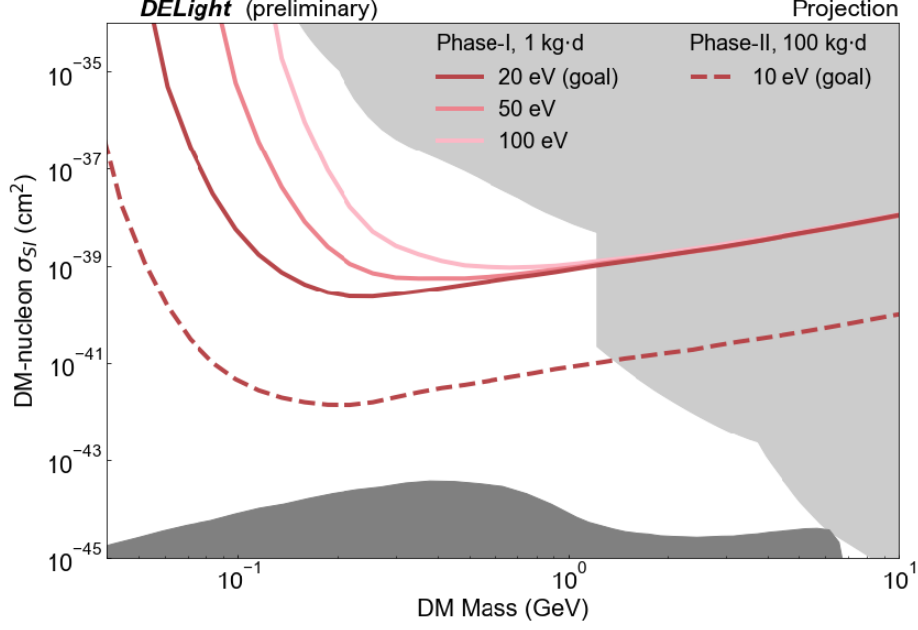


FIGURE 1. DELight’s projected sensitivity exclusion limits at different $E_{\min}$. Note that the light grey region corresponds to previously excluded regions of the parameter space. The dark grey region corresponds to the region dominated by an irreducible neutrino background. Figure adapted from [10].

1.2. Event Reconstruction for the DELight Experiment

Extracting physical parameters of recoil events from sensor traces, event reconstruction, is critical for placing constraints on LDM candidates. Placing constraints on an LDM scattering cross section, σ , for a given m_χ requires identifying the number of signal events, N_{obs} , above the energy threshold. For N_{obs} smaller than the hypothesised count of signal events, N_{exp} , for a given m_χ and σ we can place an upper bound on σ , up to some confidence level.

Discriminating between electronic recoil (ER) and potential LDM-nucleus scattering, nuclear recoil (NR) events, through ER/NR classification is critical for background rejection. LDM and neutron scattering events are characterised by NRs whilst most of the background is dominated by ERs produced from gamma ray interactions [11]. ER and NR events with recoil energies at or above the first helium excitation threshold at 19.82 eV have different signal partitioning dynamics and therefore different proportions of each of the three signal channels (see Figure 2).

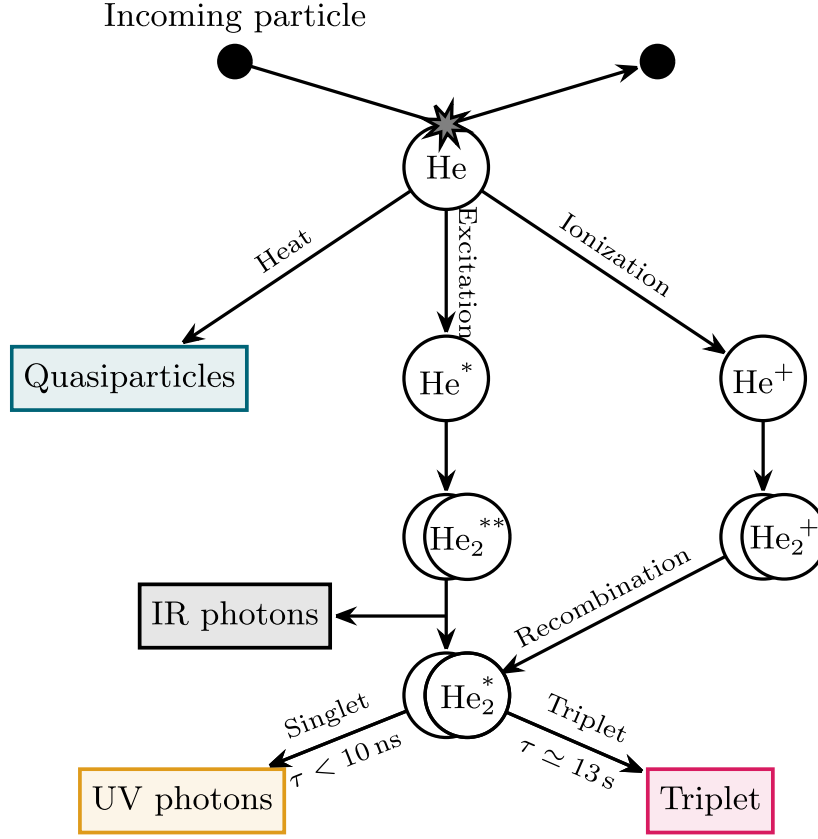


FIGURE 2. Simulation signal partitioning dynamics as outlined in [7]. Note that each of the signal channels are driven by different physical processes. Figure adapted from [7].

This difference in signal partitioning serves as a key discriminating feature for ER/NR classification and by extension background rejection [7]. For $E_R < 19.82$ eV the entirety of both ER and NR energy depositions is confined to the quasiparticle channel eliminating this feature's utility for ER/NR classification below this threshold.

Signal partitioning's dependence on E_R extends beyond the 19.82 eV excitation threshold. Quasiparticle signatures from elastic scattering dominate the signal partitioning for NR events up to 500 eV with a small but non-negligible contribution to the NR signal partitioning emerging between 100 eV and 200 eV (see Figure 3). This contribution aligns with the increasing NR excitation cross section [7]. An increase in the number of either ER or NR excitations corresponds to a larger fraction of singlets than triplets and therefore more UVs than triplet signals produced proportionately [7].

Contrastingly, the ER partitioning becomes increasingly heterogeneous for $E_R \geq 19.82$ eV. For $E_R \geq 19.82$ eV we observe the number of ER induced excitations increase with energy and by extension the number of UV and triplet signals with the former produced in a greater quantity. At $E_R = 24.59$ eV recoil events gain access to ionisation [19] by which triplets and singlets and therefore UV and triplet signals are produced with equal likelihood [20]. This therefore increases the overall fraction of triplet signals for ER events above the ionisation threshold before which UV signatures are produced in a greater fraction than triplets. For E_R at a few hundred eV and above we observe a more consistent energy partitioning for ERs [7].

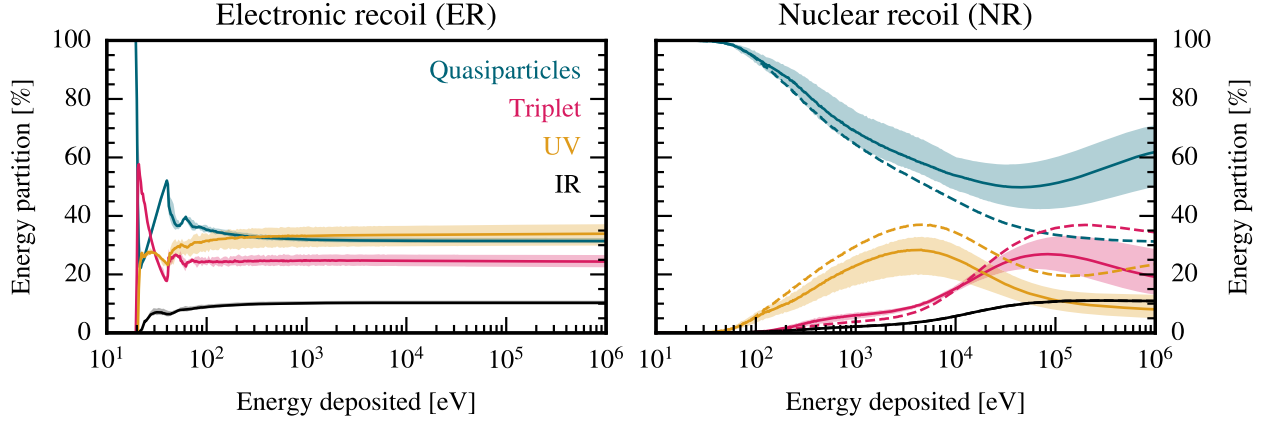


FIGURE 3. Fractional signal channel partitioning for both ER (left) and NR (right) recoil events. Note the infrared (IR) channel contributes negligibly across the 10 eV to 500 eV range so its explicit consideration is omitted for reconstruction. Figure adapted from [7].

Another principal reconstruction task necessary for background rejection is position reconstruction, predicting the location of a given recoil event in the detector. Radioactive decay products are produced near the detector walls and so DELight defines a fiducial volume [11] outside which events may be classified as background and discarded. Notably not all background events are ERs. Neutrons produced from radioactive material surrounding the detector walls can result in NRs emulating signal events [11, 21]. Position reconstruction allows for accurate identification of radioactive background events from the detector walls lying outside the fiducial volume and therefore is critical for rejecting not only ERs but also otherwise irreducible NR backgrounds.

To determine the number of observed events with $E_R \geq E_{\min}$ we must make predictions about the value of E_R for a given event, $\hat{E}_R$, from the recorded sensor data, a task known as energy reconstruction. The resolution of this energy reconstruction defines the spread of predictions about the target energy. If the energy resolution is poor, then the risk of mischaracterising events above or below $E_{\min}$ is large and so increases the risk of falsely establishing a constraint on an LDM candidate's scattering cross section. As a result, accurate energy reconstruction is critical for accurately characterising potential LDM signatures.

Maximising the performance of these three tasks that make up the foundation of DELight's event reconstruction analysis is critical for accurately placing constraints on regions of the DM parameter space and therefore forms the primary focus of the project.

1.3. Approaches to Event Reconstruction

Optimum filtering (OF) [22] is an existing approach to extracting amplitude information from trace data which has been adapted for DELight's energy reconstruction [23]. Use of OF guarantees the minimum variance estimate of a signal's amplitude under the assumptions of known and fixed noise and background templates [22]. This optimality makes the use of OF computed features passed to a boosted decision tree (BDT) [24] a natural baseline for event reconstruction. Notably however OF's optimality assumptions break down in cases where the noise profiles evolve over time and during events with mixed signal types within the same trace.

Deep learning (DL) models have seen notable adoption for DM event reconstruction with direct detection experiments including LUX-ZEPLIN [25] and XENONnT [26] having employed neural networks for pulse classification and position reconstruction. The emerging use of DL in direct detection analysis has been spurred by DL’s flexibility in learning representations across different detector geometries and reconstruction tasks.

A cross comparison between transformer neural networks [27] and BDT classifiers for the XENONnT experiment concluded that on raw data the transformer methods outperformed the BDT method for binary classification [28]. This is particularly relevant to motivating the use of transformers in DELight where event reconstruction goes beyond solely classification and presents additional challenges. DELight-specific challenges include complex signal partitioning and the scattered distribution of signal features across sensor channels and over long time series traces (see Figure 4). Transformer neural networks [27] are particularly suited to capturing intra channel and cross channel dependencies through geometrically motivated modifications to their attention mechanisms. Adapting and training transformers to perform effective event reconstruction for DELight is therefore core to the project.

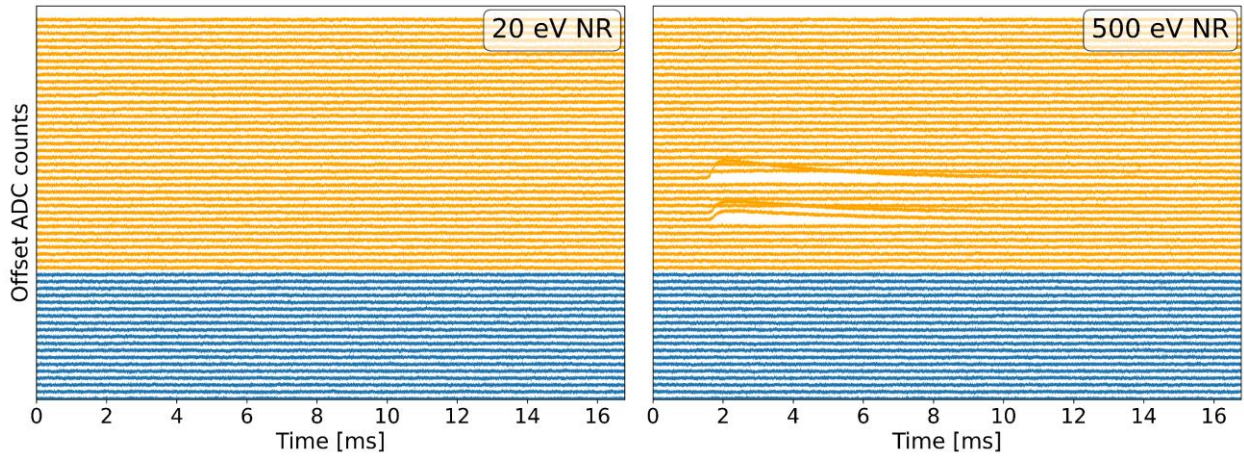


FIGURE 4. NR event traces at 20 eV and 500 eV. Orange traces denote top channel readings and blue traces bottom channel readings. Note that at 20 eV the pulse is almost entirely embedded in noise making reconstruction challenging.

1.4. Report Objectives and Overview

The work detailed in this report is motivated by three principal objectives. The first project objective is developing and training a transformer-based model [27] to perform DELight’s three event reconstruction tasks over E_R from 10-500 eV. As discussed in Section 1.2., effective event reconstruction is critical for placing constraints on regions of the DM parameter space. The second aim of this project is to evaluate the transformer model’s performance and benchmark it against an OF [23] with BDT [24], OF-BDT, baseline model. Conducting this evaluation both grounds the transformer model’s relative effectiveness and lays the foundation for exploring the energy dependence of each task’s performance. Finally, the project aims to investigate the transformer model’s capacity to learn the underlying physics of signal partitioning through a channel masking analysis. This analysis will provide insight as to whether the model is well approximating physics grounded mappings or learning spurious correlations in the data.

The structure of the report is as follows. Section 2 details the methodology underpinning the principal objectives of the report. This first includes detailing the data used to train and evaluate reconstruction models, both the transformer and its variants as well as the BDT baseline in Section 2.1 followed by an exploration of the models’ architecture details themselves in Sections 2.2-2.3. Following from this the section then discusses the approaches to training in Section 2.4 as well as the details of model evaluation and channel masking analysis in Section 2.5.

Section 3 starts with a focus on the performance of the models on all three reconstruction tasks in Sections 3.1-3.3, cross comparing the transformer model and its variant’s results against the baseline’s before concluding with a summary of the outcomes of the channel masking analysis in Section 3.4.

Finally, the results from Section 3 are contextualised and discussed in Section 4 before the conclusions of the report are presented in Section 5.

2. Methodology

2.1. Data

The foundational dataset used in the project consists of 300,000 recoil events, serving as inputs to the models, evenly split both between ERs and NRs. The events are also uniformly distributed over six recoil energies $E_R \in \{10, 20, 50, 100, 200, 500\}$ eV as well as over positions throughout the volume of the detector. The dataset is generated by the DELight collaboration using TraceSimulator [7] which simulates signal partitioning dynamics following either ER or NR events taking place in the detector.

Each event consists of one voltage trace with $T = 65,536$ timesteps for each of the $C = 56$ sensor channels. Each of the traces emulates voltage readouts sampled at $f_s = 3.91$ MHz over 16.8 ms. The events are stored in HDF5 files with 100 events batched together in each file enabling efficient file storage and data loading during training.

A noise power spectral density (PSD) is derived from noise only readings prior to the arrival of X-ray signals at an MMC [23]. This PSD is used to scale standard normally distributed complex coefficients which following an inverse Fourier transform jointly form the noise distribution for

events. These semi-empirically derived noise distributions are injected into traces additively allowing models trained on the data to better perform in realistic detector conditions. Notably the PSD serves an additional purpose in its use in OF [23] which is further discussed in Section 2.2.

When loaded the dataset is split into three subsets stratified by both recoil type and energy. These three subsets consist of a training set containing 70% of the events, a validation set containing 10% of the events and finally a holdout test set containing the remaining 20% of the dataset.

Finally, we precompute and store sensor edge features, from known sensor positions, which we inject into the model’s forward pass, detailed further in Section 2.3. Sensor edge features refer to features that encode pairwise geometric relationships between sensors in the detector. These include the 3D Cartesian displacement vector between the sensors as well as a binary flag indicating whether both sensors lie in the same plane, top or bottom, or not.

2.2. Optimum Filter and Boosted Decision Tree Baseline

The OF implementation used in this project builds on work done within the DELight collaboration based on previous applications to MMC data [23] and the foundational construction as described in [22]. OF is a powerful tool in that it yields the efficient linear estimator under assumptions of Gaussian distributed residuals as well as a known signal template, $A(f)$, and PSD, $J(f)$ [22].

The event traces are first broken down into $N_W = 64$ windows. A given signal template assumes a pulse shape beginning at the origin and therefore a larger N_W will increase the likelihood of overlap between the template and the pulse given the corresponding signal type is present in the trace. The drawback to larger N_W is that it increases the computational cost of conducting the Fourier transforms over the dataset and so N_W is chosen to balance this cost against the accuracy of the parametric fits.

Each trace window is then independently Fourier transformed giving us the corresponding frequency components $S(f)$ for each window [22, 23]. Notably these operations are performed in batches in parallel on a GPU, where otherwise the operations would be computationally prohibitive given the scale of the dataset.

For each window we obtain the amplitude, a , which maximises the likelihood of observing the observed $S(f)$ by minimising the chi-square function [22, 23]:

$$\chi^2(a) = \sum_f \frac{|S(f) - aA(f)|^2}{J(f)} \quad (1)$$

Minimising (1) with respect to a gives us [22, 23]:

$$\hat{a} = \frac{\sum_f S^*(f)A(f)/J(f)}{\sum_f |A(f)|^2/J(f)} \quad (2)$$

Where a quasiparticle signal template is applied to the top channels, and a UV signal template is applied to the bottom channels. The UV template is used for extracting both UV and triplet amplitudes as both signal types have similar pulse shapes, but distinct arrival times as discussed in Section 1.1. The parameter, $\hat{a}$, serves as the first of two channel-wise features that we pass to the

baseline BDT. Using fixed templates mapping to the known signal types, for top and bottom channels independently, the pattern of $\hat{a}$ across the signal channels allows the baseline model to determine which signal types are present during an event. Following this estimation an additional feature is extracted from each trace. Computing χ^2 at $\hat{a}$ provides context as to the goodness of fit when obtaining $\hat{a}$. This permits the model to learn whether estimated values of $\hat{a}$ with larger associated χ^2 are less reliable when making predictions.

The BDT implementation used for our baseline, XGBoost [24], is a single output model. We therefore train distinct models for each reconstruction objective including, unlike the transformer-based model discussed in Section 2.3, separate constructions for both x and y coordinate prediction during position reconstruction. For consistency however we employ the same BDT hyperparameters for each task (see Table 1).

TABLE I. XGBoost [24] baseline hyperparameters tuned empirically. Shared across all reconstruction tasks.

Hyperparameter	Value
Number of estimators	2000
Maximum tree depth	8
Learning rate	0.05
Early stopping patience	50

2.3. Transformer Model Architectures

The transformer model developed for this project (see Figure 5) is implemented in PyTorch [29] and takes events $\mathbf{X} \in \mathbb{R}^{C \times T}$ as inputs. The model jointly predicts four scalars one for ER/NR classification and energy reconstruction and two for 2D position reconstruction.

Transformer models are characterised by their ability to capture non-linear relationships between elements over long sequences through the attention mechanism [27]:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (3)$$

Where $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ are matrix groupings of query, key and value vectors respectively and d_k is the embedding dimension of said vectors. These query, key and value vectors are generated through independent linear projections of the embedded sequence. Each query vector may be viewed as an element in the sequence answering “what am I looking for?”, each key vector answering “what information can I provide?” and each value vector answering “what should be communicated?” [27]. Attention itself is a learned aggregation method defining the communication between elements in a sequence. Multi-head attention (MHA) [27] extends (3) by splitting each linearly projected vector along the model’s embedding dimension d_{model} into n_h heads such that $d_k := d_h = d_{\text{model}}/n_h$ and computing attention for each head in parallel.

We modify MHA [27] to align with the construction outlined in [30] with the addition of a sigmoid gate:

$$\text{GatedMHA}(\mathbf{X}) = \left(\text{MHA}(\mathbf{X}) \odot \text{sigmoid}(\mathbf{X}\mathbf{W}_g) \right) \mathbf{W}_o \quad (4)$$

Where $\mathbf{W}_g$ and $\mathbf{W}_o$ are the gate and output projection weight matrices respectively. The softmax operation in attention [27], through normalisation, guarantees at least some contribution from all cross-channel interactions. Through the gating in the construction (4) [30] the model can suppress interactions between sensor channels where no signal is present with the same reasoning applying to intra channel attention. Within each transformer block a SwiGLU [31] multi-layer perceptron (MLP) is implemented following GatedMHA [30].

The attention operation (3) [27] scales in memory as $\mathcal{O}(L^2)$ where L is the length of a sequence for which attention is computed over. Attention therefore over all C sensor channels and timesteps T scales as $\mathcal{O}(T^2 C^2)$ in memory which is not tractable on modern hardware. To address this our transformer model captures temporal relationships first via intra-channel attention followed then by geometric spatial relationships through cross channel attention.

Even with the sequential processing construction, naively computing full intra channel attention over $T = 65,536$ timesteps still results in storing $\approx 4.3 \times 10^9$ values in memory for the forward pass alone. We address this limitation using patched embeddings [32] whereby we first segment each trace into $N = 512$ non overlapping patches of length $P = 128$. This reduces the effective length of each trace by a factor of 128. Following this each patch is linearly projected upwards to d_{model} and as such we may view the process of patched embedding [32] as a form of learned down sampling.

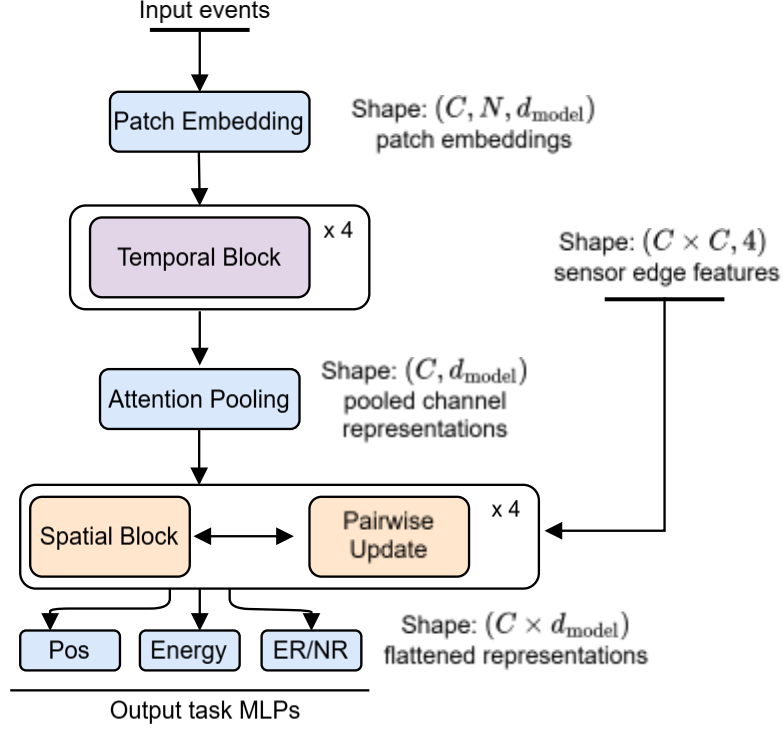


FIGURE 5. Transformer architecture overview. Input events, $\mathbf{X} \in \mathbb{R}^{C \times T}$, are temporally processed through intra-channel attention followed by spatial processing via cross-channel attention. Note the model jointly outputs predictions for all three event reconstruction tasks.

The naïve attention construction detailed in (3) [27] is permutation invariant, agnostic to the ordering of the patches in each sensor channel. This requires us to adjust the approach for temporal processing to capture the timing differences between signals and pulse shapes within sensor channels. To do this we employ Rotary Positional Embedding (RoPE) during self-attention to break this permutation symmetry [33]. We apply 2D rotation matrices $\mathbf{R}_{\theta_{i,m}}$ for $m \in \{1, \dots, d_h/2\}$ labelling the pair of key or query features being rotated by angle [33]:

$$\theta_{i,m} = i\Theta \frac{-(2m-1)}{d_h} \quad (5)$$

Where $\Theta = 10,000$ is a fixed parameter chosen in line with the literature [33] and i labels key or query's position in the sequence. The scalar product between when rotating query feature pair $\mathbf{q}_m^{(i)}$ and key feature pair $\mathbf{k}_m^{(j)}$ at positions i and j respectively is given by [33]:

$$\left(\mathbf{R}_{\theta_{i,m}} \mathbf{q}_m^{(i)}\right)^T \left(\mathbf{R}_{\theta_{j,m}} \mathbf{k}_m^{(j)}\right) = \mathbf{q}_m^{(i)T} \mathbf{R}_{\theta_{j-i,m}} \mathbf{k}_m^{(j)} \quad (6)$$

Where notably the result only depends on the relative distance, $(j - i)$, between the positions of $\mathbf{q}_m^{(i)}$ and $\mathbf{k}_m^{(j)}$ [33]. It is the relative distance between patches in the trace as opposed to their absolute position in the sequence which informs the model about both pulse shape and differences in signal timings. The choice of RoPE [33] in place of absolute positional embeddings such as those used in the original transformer paper [27] is therefore physically motivated.

Processing intra-channel relationships before cross-channel ones permits us to reduce the dimensionality of the inputs processed by the model whilst still maintaining the necessary geometry in the data required for position reconstruction. Multi-head attention pooling [34] bridges the gap between spatial and temporal processing by dynamically pooling embedded $\mathbf{X} \in \mathbb{R}^{C \times N \times d_{\text{model}}}$ representations to $\mathbf{X}' \in \mathbb{R}^{C \times d_{\text{model}}}$. This reduces the computational cost of the model by compressing each of an event's C traces to a single embedding per channel.

Following intra-channel processing the model builds and iterates on event-dependent pairwise features $\mathbf{Z} \in \mathbb{R}^{C \times C \times d_{\text{model}}}$ (see Figure 6) adapting AlphaGenome's Pair update block [35] to incorporate the detector's static sensor edge features. The sensor edge features serve as geometric priors which assist the model in encoding the positional dependence between two signatures recorded at different sensor channels. Through refinement over spatial blocks, $\mathbf{Z}$ informs cross-channel attention, allowing the model to leverage event-specific geometry when making predictions for each reconstruction task.

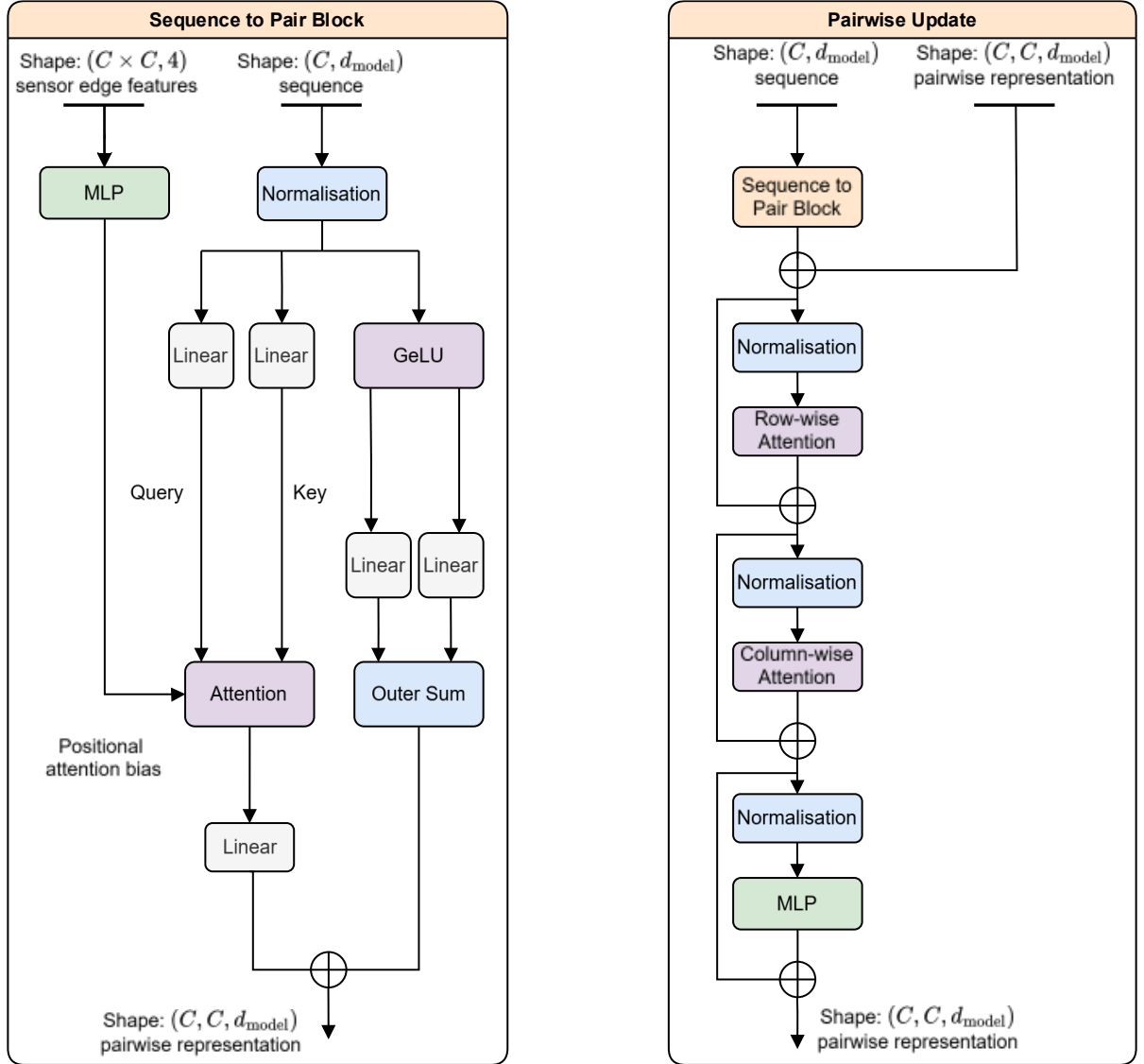


FIGURE 6. Pairwise representation construction and updates adapted from AlphaGenome [35] with the addition of column-wise attention. Note that the outputs of the Pairwise Update blocks are passed as biases during cross channel attention.

The encoding of event-specific geometry through pairwise features is particularly relevant to the task of position reconstruction where encoding the relative amplitudes and distances between event signatures is necessary to reconstruct the position of an event.

All three reconstruction tasks leverage the same shared backbone to learn representations which encode the same underlying physics required to perform all three tasks. These embeddings are passed to separate MLPs. Each MLP transforms the shared embeddings into task-specific representations and outputs predictions.

Finally, we also construct a variant of the architecture outlined in Figure 5 by injecting the OF amplitudes [22, 23] discussed in Section 2.2 into the model’s forward pass following attention pooling [34]. This is done by first projecting each of the C OF amplitudes, $\hat{a}$, per event up to d_{model} dimensional representations with an MLP and then adding them to $\mathbf{X}'$. Injecting the OF amplitudes [22, 23] provides the OF-transformer model with a strong inductive bias used to inform event reconstruction. This feature injection serves as a hybrid of traditional statistical filtering and modern DL.

2.4. Model Training

The transformer model is trained for $N_s = 10,000$ optimisation steps with an effective batch size of $B = 256$ events. Training on a single NVIDIA A100 GPU with 40 GiB of memory we are limited to loading only batches of size $B_{\text{device}} = 8$ onto the GPU at a time. To reach the target B we use a process called gradient accumulation where for each step we run $B/B_{\text{device}} = 32$ forward passes and loss calculations called micro-steps. During gradient accumulation we only update the model parameters after the final-micro step for each step. The result of each parameter update is mathematically equivalent to training with the full $B = 256$ whilst simultaneously reducing peak GPU memory usage. This peak memory reduction is a consequence of fewer intermediate values being stored each forward pass with the usage being instead amortised over the 32 micro-steps.

To train the model on all three reconstruction tasks simultaneously we use the multitask loss function:

$$\mathcal{L}_{\text{total}} = \lambda_E \mathcal{L}_E + \lambda_{\text{pos}} \mathcal{L}_{\text{pos}} + \lambda_{\text{ER/NR}} \mathcal{L}_{\text{ER/NR}} \quad (7)$$

Where the energy reconstruction loss, $\mathcal{L}_E$, is the mean squared error (MSE) between the model’s predicted energy $\hat{E}_R$ and the target energy E_R given by:

$$\mathcal{L}_E = \frac{1}{N_e} \sum_{i=1}^{N_e} (E_{R,i} - \hat{E}_{R,i})^2 \quad (8)$$

For N_e events in the dataset. The position reconstruction loss $\mathcal{L}_{\text{pos}}$ is given by the MSE between the predicted position coordinates, $\hat{\mathbf{r}}_i = (\hat{x}_i, \hat{y}_i)$ and target coordinates, $\mathbf{r}_i = (x_i, y_i)$:

$$\mathcal{L}_{\text{pos}} = \frac{1}{N_e} \sum_{i=1}^{N_e} ||\mathbf{r}_i - \hat{\mathbf{r}}_i||^2 \quad (9)$$

Finally, the ER/NR classification loss, $\mathcal{L}_{\text{ER/NR}}$ is given by the binary cross entropy (BCE) loss:

$$\mathcal{L}_{\text{ER/NR}} = -\frac{1}{N_e} \sum_{i=1}^{N_e} (c_i \ln \hat{p}_i + (1 - c_i) \ln(1 - \hat{p}_i)) \quad (10)$$

Where $c_i \in \{0, 1\}$ are binary target labels, corresponding to ER and NR labels respectively and $\hat{p}_i \in (0, 1)$ are predicted probability scores.

The scalars $\lambda_E = 10^{-4}$, $\lambda_{\text{pos}} = 10^{-3}$ and $\lambda_{\text{ER/NR}} = 1.0$ are empirically derived weights which scale each training objective such that the overall loss gradients are ≈ 1.0 . This ensures no single training objective dominates during training whilst facilitating gradient clipping with a single maximum gradient norm threshold, $\tau = 5.0$.

All model parameters are updated during training using the AdamW optimiser [36] with an initial learning rate $\eta_0 = 3 \times 10^{-4}$ and weight decay $\lambda = 4 \times 10^{-2}$. The learning rate is updated during training according to a cosine annealing with linear warmup scheduler [37, 38]. For the first 10%, $N_w = 1,000$, of optimisation steps the learning rate increases linearly from zero to η_0 , promoting early exploration of the parameter space [37]. Following this the learning rate smoothly decreases for the remaining steps with the learning rate at step t , η_t , given by [38]:

$$\eta_t = \frac{\eta_0}{2} \left(1 + \cos \left(\frac{\pi(t - N_w)}{\max(1, N_s - N_w)} \right) \right) \quad (11)$$

During model training we mask channel inputs randomly. Each event has all top channel inputs masked with probability $p_{\text{top}} = 0.1$, all bottom inputs masked with probability $p_{\text{bot}} = 0.1$ or no channels masked with probability $p_{\text{none}} = 0.8$ where the masking configurations are mutually exclusive. Training the model with random channel masking allows it to generalise better to events with similarly missing channels. This robustness is necessary to prevent an artificial collapse in performance during our masking analysis when the model is passed inputs that would otherwise lie outside its training distribution. The effect on training is pronounced with training losses oscillating partly because of the stochastic nature of the channel masking. We see improving performance across all tasks on the validation set, suggesting that the masking has not inhibited the model's ability to generalise (see Figure 7).

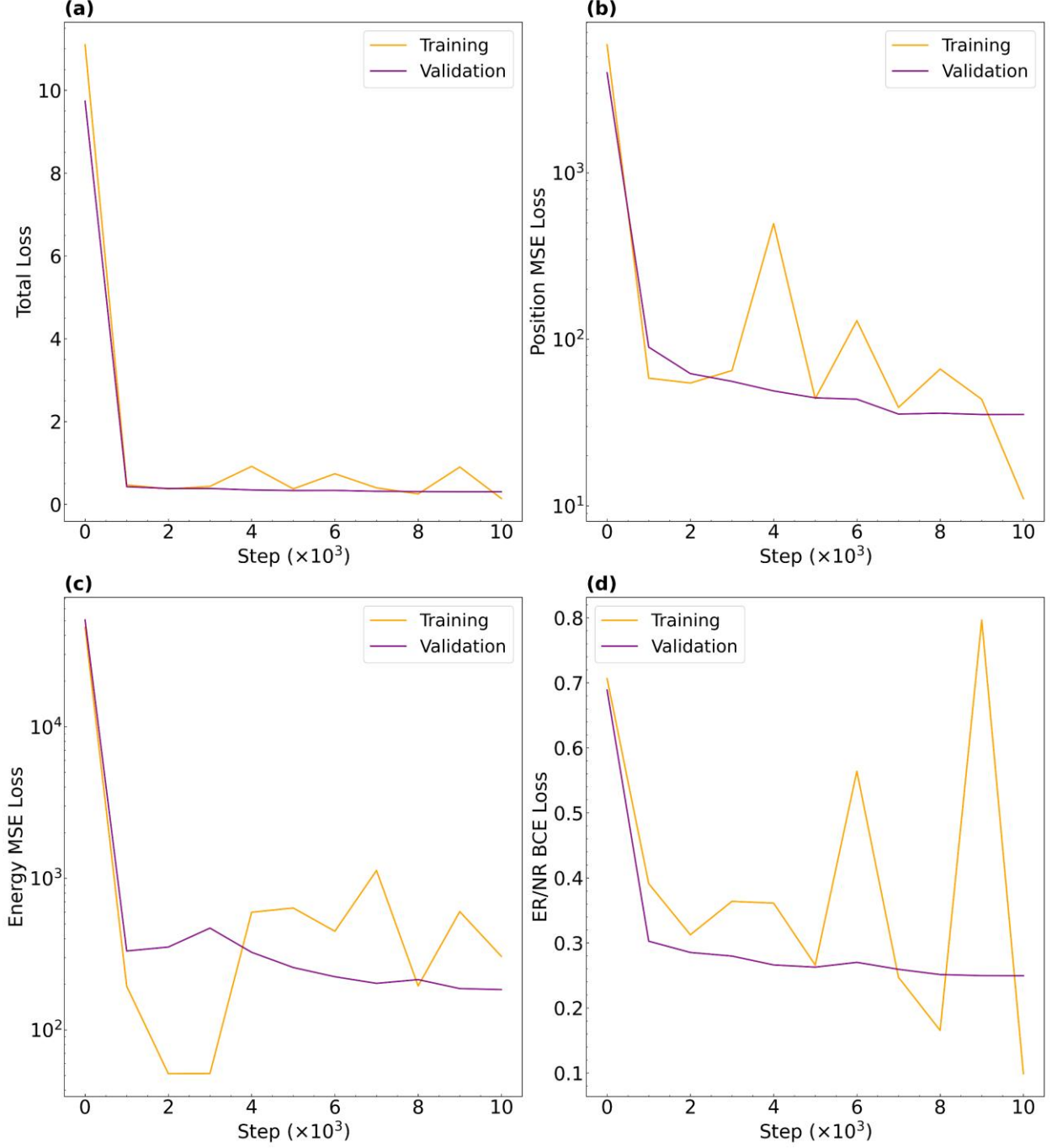


FIGURE 7. OF-transformer training and validation losses. Note that whilst the training losses oscillate wildly over each reconstruction task the validation loss plateaus and decreases steadily.

2.5. Model Evaluation and Analysis

Following training we evaluate the transformer, OF-transformer, and OF-BDT baseline's performance on the same test dataset binned by target recoil energy E_R as well as over the entire test dataset. Evaluating the models on the binned subsets allows us to compare changes in the models' performance with the energy dependence of the simulation's signal partitioning described in [7].

Our primary evaluation metric for energy reconstruction is the energy resolution σ_E/E_R where σ_E is the standard deviation of a model's energy predictions, $\hat{E}_R$ for target energy E_R . This provides a scale invariant measure of the relative spread of the model's prediction.

Additionally, to understand whether the model systematically under or over predicts for each energy bin we evaluate relative energy bias:

$$\text{Bias} = \frac{\langle \hat{E}_R - E_R \rangle}{E_R} \quad (12)$$

Where a large positive bias for a given E_R indicates a model is systematically over predicting, a large negative bias indicates systematic underprediction and a small bias suggests the evaluated model is neither overpredicting nor underpredicting.

For position reconstruction our primary evaluation metric is the root mean squared error (RMSE) between $\hat{\mathbf{r}}_i$ and $\mathbf{r}_i$ given by:

$$\text{RMSE}_{\text{pos}} = \sqrt{\frac{1}{N_e} \sum_{i=1}^{N_e} \|\mathbf{r}_i - \hat{\mathbf{r}}_i\|^2} \quad (13)$$

RMSE_{pos} penalises large outliers through squaring the difference between the predicted and target coordinates. This makes it particularly appropriate for our position reconstruction task as it is these larger outliers that are more likely to result in falsely characterising whether an event lies within the detector's fiducial volume. Additionally, unlike the MSE used for $\mathcal{L}_{\text{pos}}$ we can state RMSE in units of distance, millimetres in our case, make it more physically interpretable than MSE for evaluating model performance.

Position reconstruction is additionally evaluated using the coefficients of determination:

$$R^2(x, y) = (R_x^2, R_y^2) = \left(1 - \frac{\sum_{i=1}^{N_e} (x_i - \hat{x}_i)^2}{\sum_{i=1}^{N_e} (x_i - \langle x \rangle)^2}, 1 - \frac{\sum_{i=1}^{N_e} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N_e} (y_i - \langle y \rangle)^2} \right) \quad (14)$$

Where R_x^2 and R_y^2 are the coefficients of determination for the $\hat{x}$ and $\hat{y}$ predictions respectively, $\langle x \rangle$ is the sample mean of event x coordinates over the dataset and similarly $\langle y \rangle$ is the sample mean of event y coordinates over the dataset. Each coefficient corresponds to the proportion of the variance in the data explained by a model. A model evaluated to have $R^2(x, y) = (0, 0)$ on the dataset has performance equivalent to predicting $\langle x \rangle$ and $\langle y \rangle$ for each event. Conversely, a model evaluated to have $R^2(x, y) = (1, 1)$ on the dataset corresponds to a perfect predictor with no deviation from the true coordinates for any of the evaluated events.

For ER/NR classification we define the NR efficiency, ϵ_{NR} , as the fraction of NR events identified. Similarly, we also define the ER leakage fraction f_{ER} as the fraction of ER events incorrectly identified as NR events. For model comparison we choose ROC-AUC as our primary ER/NR classification performance metric. ROC-AUC provides an aggregate measure of a model's ϵ_{NR}

over a range of f_{ER} thresholds and is therefore a strong indicator of a model’s ability to discriminate between ER and NR events. An ROC-AUC of 0.5 corresponds to random guessing whilst an ROC-AUC of 1.0 corresponds to a perfect classifier over all the evaluated f_{ER} . We also evaluate the f_{ER} for a fixed $\epsilon_{\text{NR}} = 0.5$, representative of practical analysis conditions in line with direct detection convention [39].

The metrics ROC-AUC and RMSE_{pos} are evaluated on the entire dataset and averaged, in addition to on the binned subsets, providing a summative result of each model’s ER/NR classification and position reconstruction performance respectively. The resolution, σ_E/E_R , is only evaluated on the binned subsets as it requires a defined target energy, E_R , in its construction.

To gauge the precision of each metric calculation we use non-parametric bootstrap resampling [40]. Non-parametric bootstrap resampling [40] is conducted by first uniformly sampling several events equivalent to the energy bin size from the corresponding test dataset bin with replacement. This is then repeated 1000 times and the sample standard deviation, σ_B , is computed for each of the performance metrics at each energy bin. Consequently, our quoted error bars are $\sigma_B = \pm 1$ providing an approximate measure of the given metric’s precision assuming events sampled from the test set distribution.

Finally, to better understand whether the model has learned the underlying signal partitioning dynamics, we conduct a channel masking analysis. We conduct this by evaluating the transformer and OF-transformer’s performance as described above for the mutually exclusive cases of all top channel inputs masked, all bottom channel inputs masked, and no channel inputs masked. For the case of the OF-transformer we ensure that the OF amplitudes [22, 23] injected mid forward pass are also masked when appropriate to prevent information leakage.

3. Results

3.1. Energy Reconstruction

The OF-transformer model achieves a lower σ/E_R than both the transformer and the OF-BDT baseline across all E_R (see Figure 8, Table 2) on the test dataset. At $E_R = 20$ eV the OF-transformer delivers a resolution of $\sigma/E_R = 17.1\%$ against the OF-BDT baseline result of $\sigma/E_R = 20.8\%$ demonstrating the OF-transformer’s advantage at DELight’s Phase I target detection threshold [11]. The OF-transformer achieves the lowest resolution at $E_R = 500$ eV, $\sigma/E_R = 2.61\%$, representing consistent energy reconstruction performance across energy scales.

All three models maintain similar σ/E_R at $E_R = 10$ eV with both the OF-transformer and OF-BDT models demonstrating a marginal improvement over the transformer (see Figure 8). Similarly, all three models demonstrate a degradation in σ/E_R from 10 eV to 50 eV. The OF-BDT resolution then diverges from that of both transformer variants for $E_R > 50$ eV. The OF-BDT performance continues to decrease from 50 eV to 100 eV before beginning to reconverge with the transformer variants from 200 eV to 500 eV.

We also observe that the OF-BDT obtains a Bias of 46.7% at $E_R = 10$ eV suggesting systematic overprediction (see Table 2). This is notably larger than both the OF-transformer Bias of 6.31% and the transformer Bias of -6.43% .

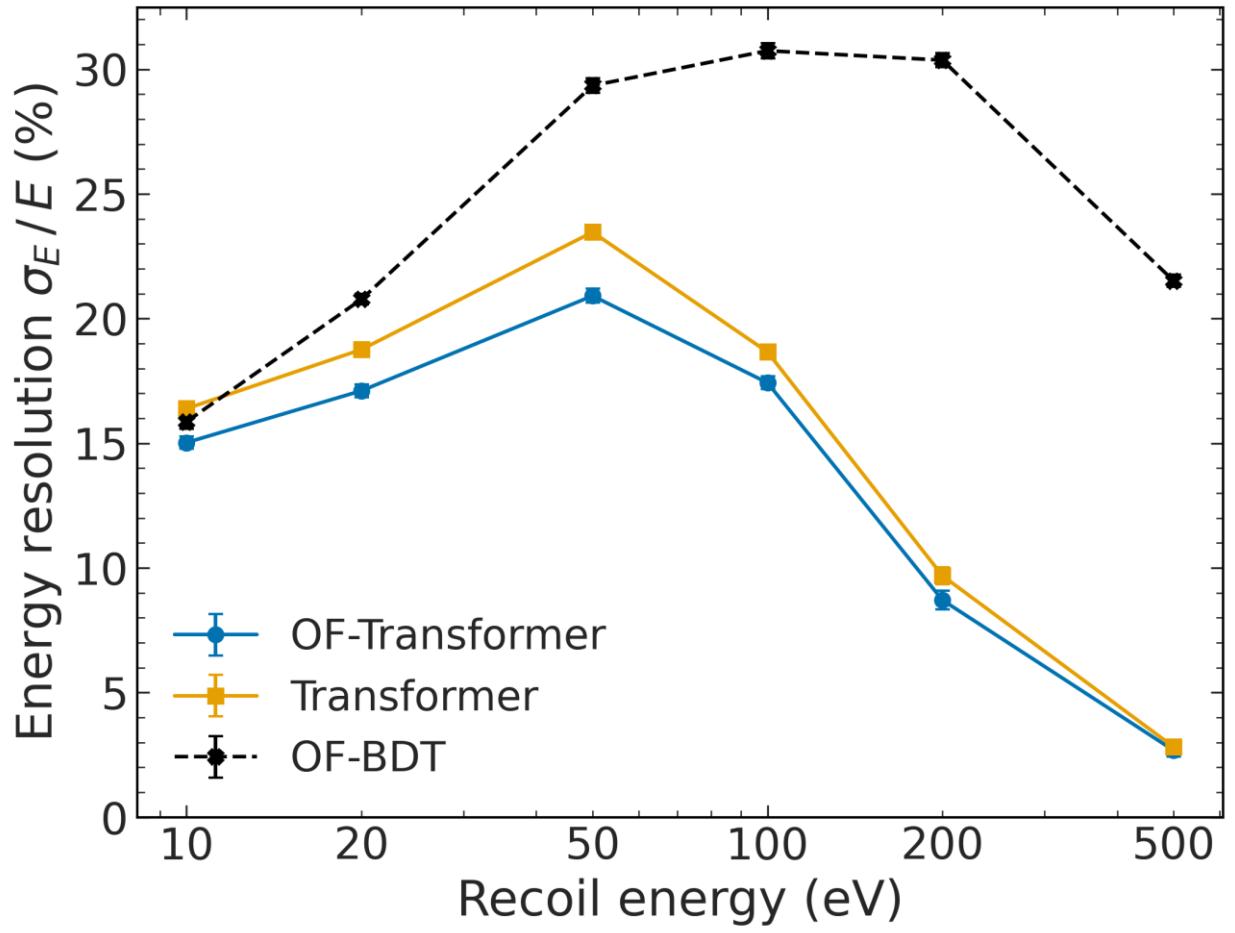


FIGURE 8. Energy reconstruction resolution across evaluated models. Resolution diverges notably between the transformer variants and the OF-BDT baseline for $E_R > 50$ eV. Also note that the OF-Transformer achieves lower resolution than the transformer for all E_R .

TABLE II. Energy reconstruction performance for each model evaluated on 10,000 events per energy bin. Best result for each metric and each energy bin is highlighted in bold.

E_R	Transformer		OF-Transformer		OF-BDT	
	σ/E_R (%)	Bias (%)	σ/E_R (%)	Bias (%)	σ/E_R (%)	Bias (%)
10 eV	16.4	6.31	15.0	-6.43	15.9	46.7
20 eV	18.8	4.02	17.1	-0.694	20.8	0.134
50 eV	23.5	5.67	20.9	3.66	29.3	-14.0
100 eV	18.7	2.35	17.4	3.73	30.7	-4.90
200 eV	9.69	4.20	8.71	2.67	30.4	4.20
500 eV	2.82	2.57	2.68	2.31	21.5	-2.30

3.2. Position Reconstruction

We evaluate the models on the RMSE_{pos} metric and find that the OF-transformer outperforms both the transformer and OF-BDT baseline across all E_R aside from $E_R = 20$ eV (see Figure 9, Table 3). At 20 eV the native transformer demonstrates a marginally lower $\text{RMSE}_{\text{pos}} = 11.3$ mm relative to the OF-transformer’s $\text{RMSE}_{\text{pos}} = 11.5$ mm (see Table 3). Both transformers demonstrate substantially improved RMSE_{pos} at this E_R relative to the OF-BDT baseline which obtains $\text{RMSE}_{\text{pos}} = 33.9$ mm. The OF-transformer obtains the lowest RMSE_{pos} at $E_R = 500$ eV representing consistent position reconstruction performance at both low and high E_R .

Both transformer variants demonstrate a decrease in position reconstruction performance from 10 eV to 20 eV (see Figure 9). This temporary performance decrease contrasts with the OF-BDT which sees its RMSE_{pos} monotonically decrease across the energy range.

We observe that all three models achieve $R^2(x, y)$ values greater than 0.9 reflecting a uniform capacity to capture the variance in the dataset. Finally, we also observe that in all three cases R_x^2 lies within $\sim 0.5\%$ of R_y^2 demonstrating a lack of coordinate dependent asymmetry in position reconstruction performance.

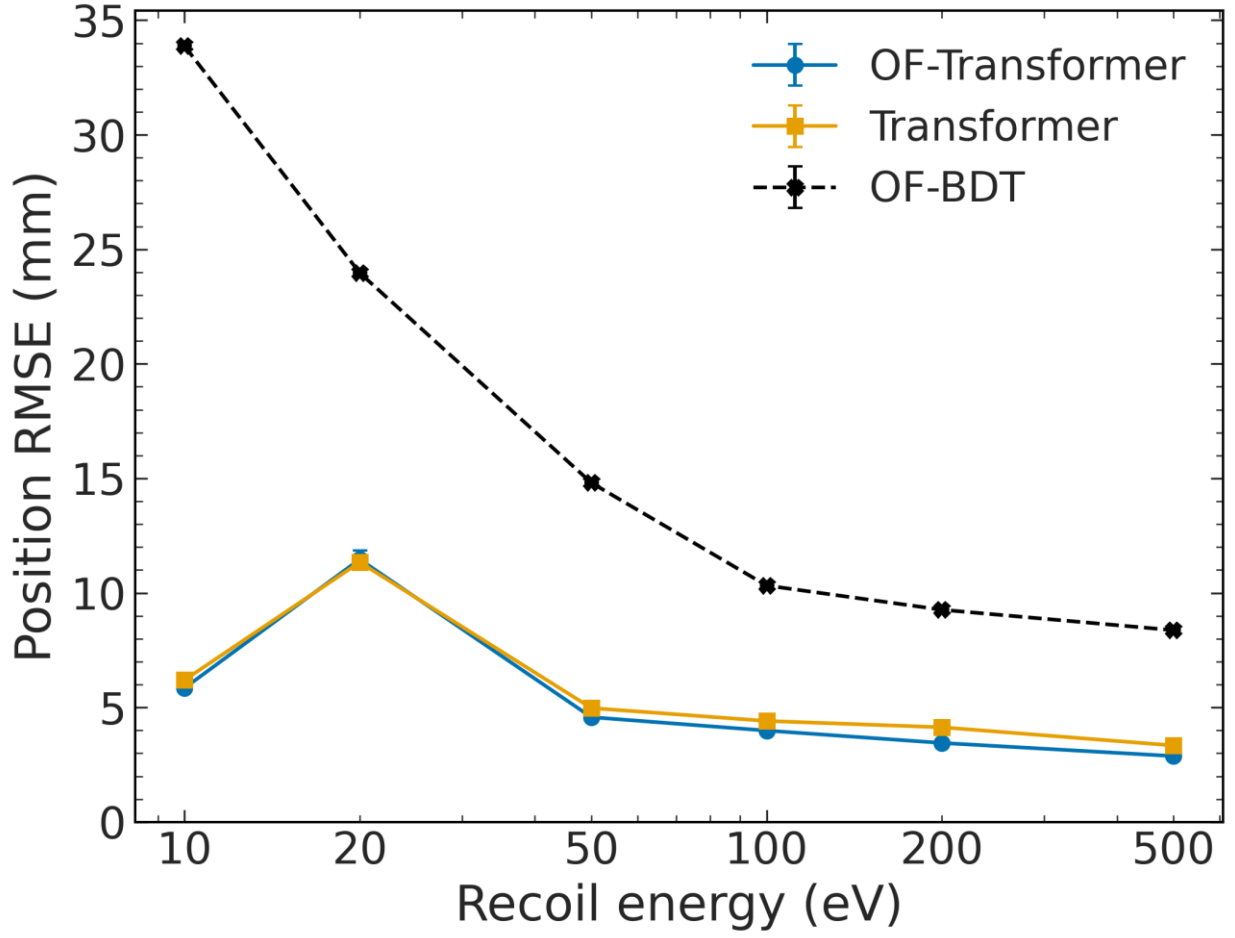


FIGURE 9. Position reconstruction RMSE_{pos} across evaluated models. Transformer variants outperform the OF-BDT across all E_R . Note the increase in RMSE_{pos} for both transformer variants at $E_R = 20$ eV whilst the OF-BDT performance increases monotonically with E_R .

TABLE III. Position reconstruction performance for each model evaluated on 10,000 events per energy bin and on the entire test dataset of 60,000 events. Best result for each metric and each energy bin is highlighted in bold.

E_R	Transformer		OF-Transformer		OF-BDT	
	RMSE (mm)	$R^2(x, y)$	RMSE (mm)	$R^2(x, y)$	RMSE (mm)	$R^2(x, y)$
10 eV	6.20	-	5.85	-	33.9	-
20 eV	11.3	-	11.5	-	24.0	-
50 eV	4.97	-	4.57	-	14.8	-
100 eV	4.41	-	3.99	-	10.3	-
200 eV	4.13	-	3.45	-	9.26	-
500 eV	3.34	-	2.24	-	8.37	-
Overall	6.32	(0.990, 0.990)	6.09	(0.992, 0.991)	19.2	(0.908, 0.909)

3.3. ER/NR Classification

Evaluating the models on the ROC-AUC metric we find that the OF-transformer demonstrates improved performance over the OF-BDT baseline across all E_R (see Figure 10, Table 4). The OF-transformer additionally obtains ROC-AUC = 0.613 (see Table 4) at the Phase I detection threshold [11], higher not only than the OF-BDT baseline but also the transformer which achieves a marginally lower score of ROC-AUC = 0.611. All three models demonstrate poor performance at $E_R = 10$ eV with ROC-AUC ≈ 0.5 consistent with a uniform random classifier.

All three models achieve ROC-AUC > 0.9 for $E_R \geq 50$ eV, with ROC-AUC increasing rapidly from 20 eV to 50 eV (see Figure 10). For the transformer variants, in contrast to the OF-BDT, we observe a plateau in ROC-AUC in the $E_R \geq 50$ eV region. This is consistent with the f_{ER} at $\epsilon_{NR} = 0.5$ for both transformers where both achieve $f_{ER} \leq 10^{-3}$ for $E_R \geq 50$ eV highlighting almost perfect ER/NR classification performance at higher energies.

We observe an overall ER/NR classification performance advantage over the OF-BDT across the entire spectrum of ϵ_{NR} thresholds when evaluated across the entire test dataset (see Figure 11). When we establish the $\epsilon_{NR} = 0.5$ threshold for the $E_R = 20$ eV subset however we note that whilst the transformer variants achieve a higher ROC-AUC than the OF-BDT the OF-BDT obtains a lower f_{ER} than both the transformers.

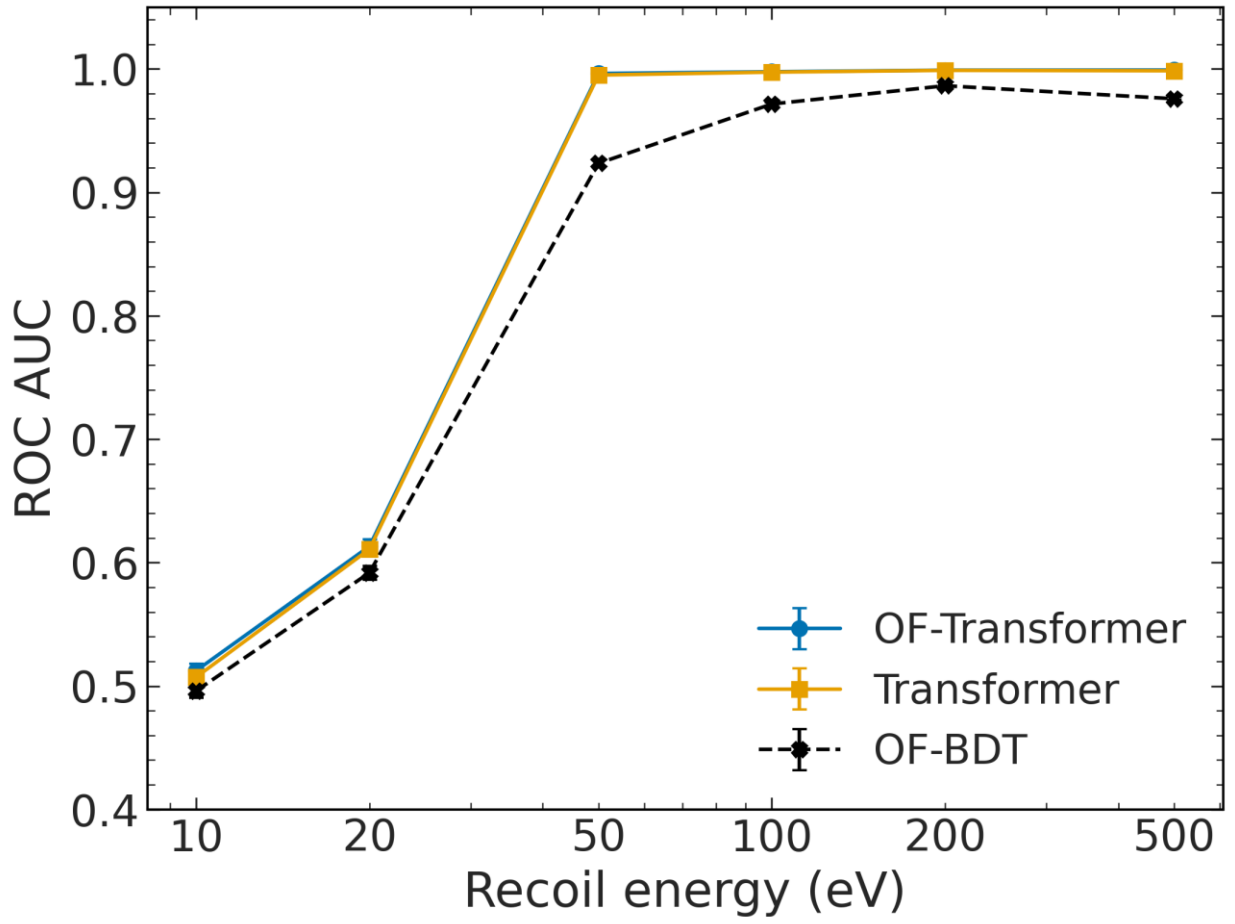


FIGURE 10. ER/NR classification ROC-AUC across evaluated models. Transformer variants outperform the OF-BDT across all E_R . Note the performance of both transformer variants begins to plateau for $E_R \geq 50$ eV.

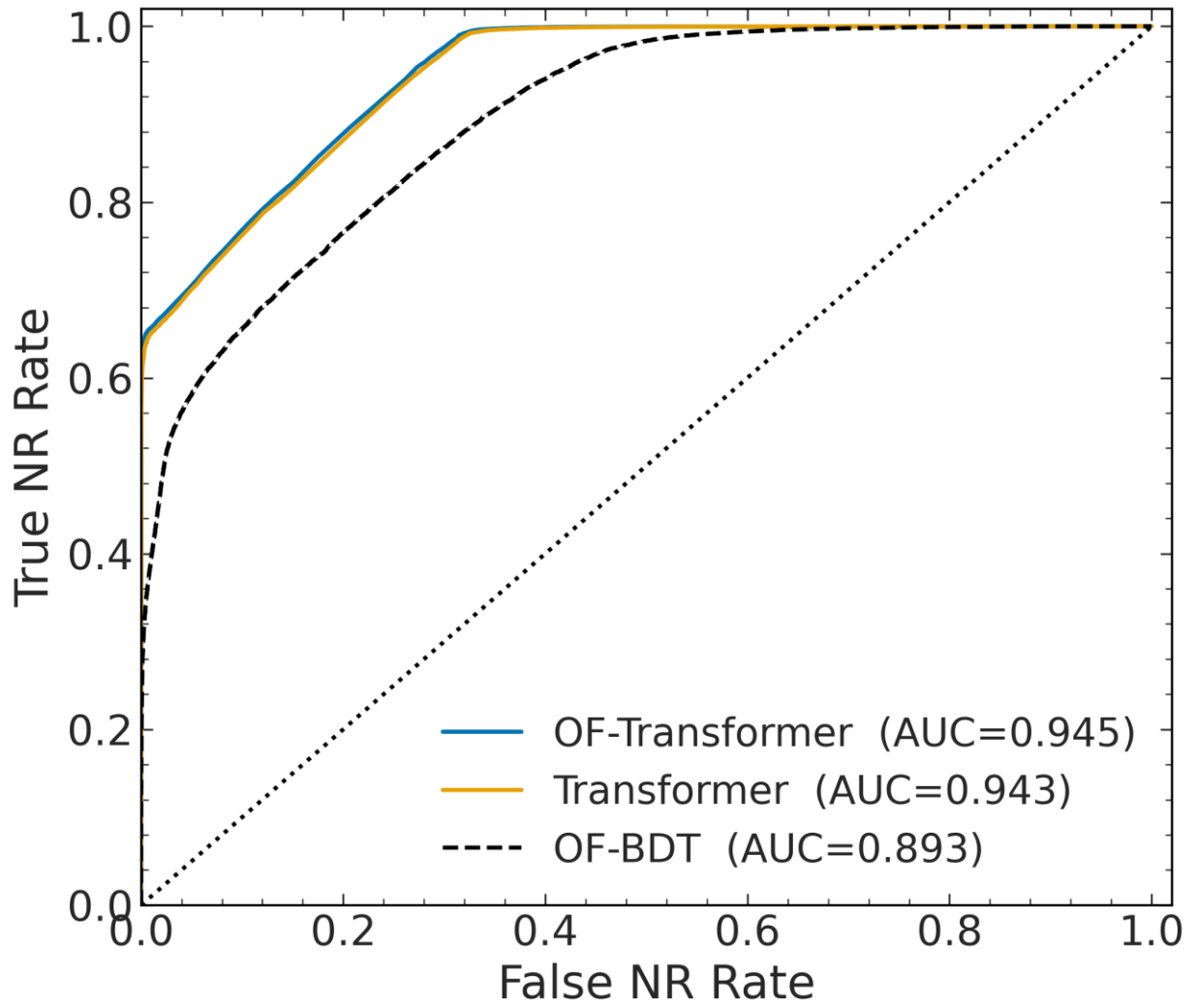


FIGURE 11. ER/NR classification ROC curves across evaluated models averaged over all E_R . Transformer variants outperform the OF-BDT across all discrimination thresholds. Note the dotted line indicates a uniform random classifier with an ROC-AUC of 0.5.

TABLE IV. ER/NR classification performance for each model evaluated on 10,000 events per energy bin and on the entire test dataset of 60,000 events. Best result for each metric and each energy bin is highlighted in bold. Note that a larger ROC-AUC and a lower f_{ER} at $\epsilon_{\text{NR}} = 0.5$ indicate improved classification performance. Overall f_{ER} at $\epsilon_{\text{NR}} = 0.5$ is not reported due to bins at $E_{\text{R}} \geq 50$ eV dominating aggregate performance.

E_{R}	Transformer		OF-Transformer		OF-BDT	
	ROC-AUC	f_{ER} at $\epsilon_{\text{NR}} = 0.5$	ROC-AUC	f_{ER} at $\epsilon_{\text{NR}} = 0.5$	ROC-AUC	f_{ER} at $\epsilon_{\text{NR}} = 0.5$
10 eV	0.507	0.517	0.512	0.579	0.496	0.503
20 eV	0.611	0.414	0.613	0.412	0.592	0.393
50 eV	0.995	0.000	0.996	0.000	0.924	0.049
100 eV	0.997	0.001	0.998	0.001	0.972	0.005
200 eV	0.999	0.000	0.999	0.000	0.986	0.000
500 eV	0.999	0.000	0.999	0.000	0.976	0.001
Overall	0.943	-	0.945	-	0.893	-

3.4. Model Analysis Results

Conducting a channel masking analysis on the OF-transformer we consistently observe the greatest performance degradation with top channel masking across all reconstruction tasks and E_{R} except for ER/NR classification at $E_{\text{R}} = 100$ eV where we observe the greatest degradation when masking bottom sensor channel inputs (see Figure 12, Table 5).

Evaluating the OF-transformer model on the σ/E_{R} metric we observe that all three configurations demonstrate an increase in σ/E_{R} from 10 eV to 50 eV with 50 eV also corresponding to the largest gap in σ/E_{R} between the masking configurations with the top channel masking resolution exceeding 100% at 50 eV (see Figure 12). From 50 eV to 500 eV however the resolutions for the three configurations begin to converge.

Both the unmasked and bottom masked sensor channel configurations demonstrate a temporary increase in RMSE_{pos} at $E_{\text{R}} = 20$ eV. This is in contrast with the top masked configuration where RMSE_{pos} decreases monotonically across the energy range. We also observe that at $E_{\text{R}} = 100$ eV there is an isolated case where bottom channel masking results in an improvement over the unmasked configuration's position reconstruction performance. The largest decrease in RMSE_{pos} occurs during top channel masking at $E_{\text{R}} = 10$ eV. In this case the RMSE_{pos} increases by a factor of ~ 10.9 . This recoil energy also yields an improved position reconstruction performance for the bottom masked configuration than with no masking at all.

All three masking configurations demonstrate similarly poor classification performance at $E_{\text{R}} = 10$ eV with $\text{ROC-AUC} \approx 0.5$. At $E_{\text{R}} = 20$ eV both the bottom and unmasked configurations obtain

ROC-AUC = 0.613 (see Table 5) with only the top sensor masking resulting in a lower ROC-AUC = 0.514. The largest decrease in classification performance however is observed during top channel masking at $E_R = 50$ eV. At this E_R ROC-AUC decreases from 0.996 when unmasked to 0.814 when top sensor channels are masked. Finally, we also note that uniquely the bottom masking configuration demonstrates a similar scale drop in ROC-AUC at $E_R = 100$ eV.

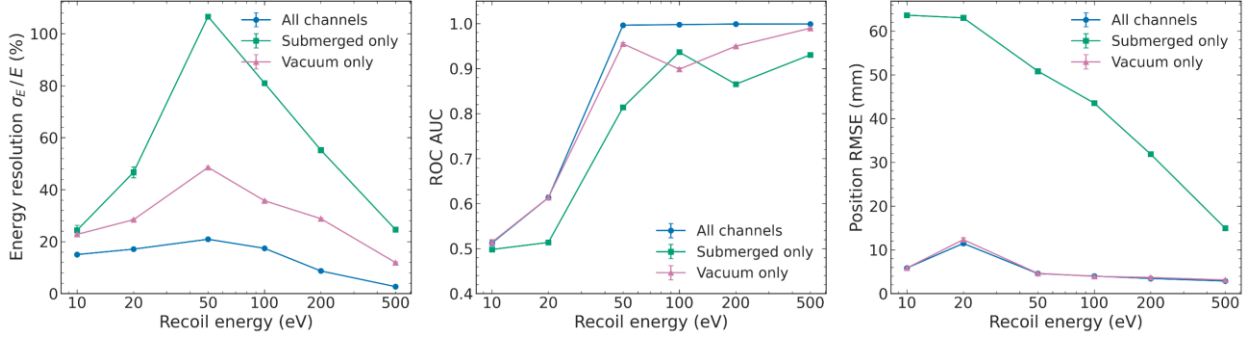


FIGURE 12. OF-transformer reconstruction performance for different sensor channel masking configurations across all three event reconstruction tasks. Masking configurations consist of no channels masked (all channels), top channels masked (submerged only) and bottom sensor channels masked (vacuum only). Note that for position and energy reconstruction top channel masking contributes the most to performance degradation across all energy scales.

TABLE V. OF-transformer event reconstruction performance for top, bottom and unmasked configurations. Model is evaluated on 10,000 events per energy bin and on the entire test dataset of 60,000 events. Best result for each metric and each energy bin is highlighted in bold.

E_R	No Mask			Top Mask			Bottom Mask		
	σ/E_R (%)	RMSE (mm)	ROC-AUC	σ/E_R (%)	RMSE (mm)	ROC-AUC	σ/E_R (%)	RMSE (mm)	ROC-AUC
10 eV	15.0	5.85	0.512	24.4	63.6	0.498	22.8	3.65	0.514
20 eV	17.1	11.5	0.613	46.6	63.0	0.514	28.4	12.3	0.613
50 eV	20.9	4.57	0.996	107	50.8	0.814	48.6	4.64	0.955
100 eV	17.4	3.99	0.998	81.0	43.5	0.936	35.7	3.95	0.899
200 eV	8.71	3.45	0.999	55.2	31.9	0.865	28.8	3.68	0.950
500 eV	2.68	2.24	0.999	24.7	14.9	0.931	12.0	3.13	0.990
Overall	-	6.09	0.945	-	47.9	0.734	-	6.41	0.854

4. Discussion

4.1. Model Comparison

The OF-transformer achieves improved σ/E_R , RMSE_{pos} and ROC-AUC performance than the OF-BDT baseline across all recoil energies. We observe the largest energy reconstruction gap between the two models at $E_R = 500$ eV. At 500 eV top sensor channels have the highest likelihood of recording UV and triplet signals in place or alongside quasiparticle signatures [7]. This is a consequence of the consistent availability of ER excitations, NR excitations and ER ionisations at this energy [7]. The OF amplitudes, $\hat{a}$, are estimated with the assumption of quasiparticle signatures in the top sensor channels and UV or triplet in the bottom sensor channels [23]. Consequently, the OF assumption of a known PSD [23] is more likely to break down at 500 eV resulting in the extracted OF amplitudes [22, 23] being more often poorly calibrated. The transformer variants, through processing the raw trace representations, better adapt to the presence of different top sensor channel signatures than the OF-BDT which has access to only OF extracted summary features. This is demonstrated by both transformer variants achieving comparably lower σ/E_R than the OF-BDT at 500 eV (see Figure 8). We also determine that the OF-transformer can leverage the OF amplitude optimality [22] in the cases of homogeneous quasiparticle signature presence in the top sensor channels as evidenced by the OF-transformer’s lower resolution than the native transformer.

Whilst the OF optimality assumptions frequently break down at higher energies, we observe the converse at $E_R = 10$ eV. At 10 eV the entirety of the signal is formed by quasiparticle signature [7] and is recorded by the top sensor channels [10]. This ensures the top sensor channel signatures are homogeneous and the OF filters are well matched to the top sensor signals. The result is that both the OF-BDT and the OF-transformer achieve lower σ/E_R at 10 eV than the native transformer whilst at all other E_R , where excitations are available and top channel signals are sometimes heterogeneous, both transformer variants outperform the OF-BDT baseline. Despite both the OF-BDT and the OF-transformer taking $\hat{a}$ as inputs the OF-transformer achieves a marginally lower σ/E_R at 10 eV. Whilst the estimator for each window’s $\hat{a}$ within each trace is unbiased, taking the maximum of these amplitudes for a given trace results in positively biased predictions when the noise magnitude is comparable to that of the signal [41]. The OF-BDT has access only to these OF extracted features and so the systematic overprediction propagates into its σ/E_R as evidenced by the OF-BDT’s larger Bias relative to both transformer variants (see Table 2). The OF-transformer on the other hand can leverage $\hat{a}$ whilst simultaneously correcting for bias through access to raw trace information resulting in lower σ/E_R at 10 eV.

From 10 eV to 50 eV all three models demonstrate an increase in σ/E_R in line with the increasing signal partitioning variance (see Figure 3) [7]. The intrinsic variance in which signal types are present in each sensor channel up to 50 eV means that given identical E_R and initial conditions two events are likely to have different amplitude profiles. This reduces the effective size of the 50 eV subset during the model training in that the model would require training on additional 50 eV events. We therefore observe all three models’, independent of architecture, energy reconstruction performance decrease up to 50 eV. In contrast to the transformer variants we observe a continued increase in σ/E_R for the OF-BDT up to 100 eV demonstrating that the transformer has learned more informative representations to better adapt to partitioning variance.

A consistent gap is observed between the position reconstruction performance of both transformer variants and that of the OF-BDT baseline across the energy range with the largest gap being demonstrated at $E_R = 10$ eV (see Figure 9). At 10 eV the OF-BDT has access only to potentially biased $\hat{a}$ estimates and the associated χ^2 values. On the other hand, the transformer variants can additionally leverage pulse shape and timing information through RoPE [33] which, alongside prior information about the detector geometry, is built into pairwise representations [35]. This difference in representational capacity at 10 eV emerges as the largest RMSE_{pos} gap between the transformers and the OF-BDT. The gap then shrinks as the OF-BDT becomes able to triangulate the event position increasingly well at higher energies where $\hat{a}$ are decreasingly derived from noisy channels.

Whilst the OF-BDT demonstrates a monotonic decrease in its RMSE_{pos} with increasing E_R as the signal to noise ratio improves, both transformer variants experience a temporary increase in RMSE_{pos} at $E_R = 20$ eV. At 20 eV ER events have access to a single possible excitation with a corresponding scintillation signal, either UV or triplet [7]. The presence of the scintillation signal is not guaranteed however and may occur either in the top or bottom sensor channels for a given event. This implies that whilst at $E_R = 10$ eV the transformer models can learn to model solely top inter channel relationships, at 20 eV there is a high degree of irreducible variance in whether any signal occurs in the top or bottom sensor plane. At higher E_R larger quantities of scintillation from ionisation and excitation are present [7] and so the transformers can learn from more consistent cross channel correlations in the aggregate. The OF-BDT on the other hand has a more limited capacity to learn these cross-channel correlations in the first place and so sees no increase in RMSE_{pos} at $E_R = 20$ eV.

All three models are evaluated to align approximately with a uniform random classifier for ER/NR classification at $E_R = 10$ eV. The dominant discriminating feature for recoil type discrimination for DELight’s construction is the ratio between the three signal types across the sensor channels [7]. As discussed in Section 1.2 below the first helium excitation threshold at $E_R = 19.82$ eV the entirety of the signal lies in the quasiparticle channel for both ERs and NRs [7]. Consequently, the poor performance at $E_R = 10$ eV for all three models is consistent with the signal partitioning physics.

At $E_R = 20$ eV, whilst both transformer variants demonstrate improved ROC-AUC over the OF-BDT baseline the OF-BDT baseline achieves a lower f_{ER} at $\epsilon_{\text{NR}} = 0.5$ (see Table 4). This indicates that the transformers are better at classifying the small fraction of events at 20 eV where scintillation is produced [7] and so would achieve an improved f_{ER} over the OF-BDT at a lower ϵ_{NR} threshold where more of the indeterminable events, those with only quasiparticle signature, are rejected. Across all other recoil energies both transformers outperform the OF-BDT baseline with the OF-transformer achieving consistently marginally improved ROC-AUC over the native transformer afforded by the OF-transformer leveraging $\hat{a}$. The ROC-AUC of both transformer variants saturates at higher energies (see Figure 10) in line with the stabilisation of the signal channel partitioning (see Figure 3) [7]. This saturation does not occur to a similar extent with the OF-BDT where the model is limited by errors in the amplitude estimation without the transformers’ capacity to correct using the trace representations.

Transformers outperform the OF-BDT across the majority of E_R and all three reconstruction tasks by extracting pulse timing and shape information through RoPE [33] and then compressing the encoded features into richer channel representations with MHA pooling [34]. The pooled

representations improve energy reconstruction performance in the OF-transformer by facilitating corrections to noisy amplitude estimates. Additionally, through iterative refinement of the pairwise representations [35] the transformers can project from one dimensional, 56 channel, sequences to capturing three-dimensional cross sensor correlations. This approach affords the transformers a marked improvement in position reconstruction over the OF-BDT which is limited to an assumption of channel independence through its architecture. Without the limitation of static filters [22, 23] the transformers better adapt to identifying heterogeneous signal types within top sensor channels than the OF-BDT. Accurate identification of signal types results in better capacity for characterising the fractional signal partitioning. Consequently, the transformers obtain better ER/NR classification performance at higher energies.

4.2. Channel Masking Comparison

Energy reconstruction performance degrades most substantially when top sensor channel inputs are masked across all E_R . Both top and bottom sensor channels can record scintillation signal produced by events when available, however only top sensor channels can record quasiparticle signature [10] which dominates at lower E_R [7]. As a result, a greater fraction of amplitude information will lie in the top sensor channels given uniformly distributed event positions as is the case with our dataset (see Section 2.1). Consequently, observing the greatest increase in σ/E_R when masking top sensor channels is physically consistent with the distribution of amplitude information between the sensor channels.

The largest degradation from top sensor channel masking is observed at $E_R = 50$ eV. This recoil energy is that where we observe the largest signal partitioning variance (see Section 4.1) [7]. As a result, removing most of the available amplitude information, most of which lies in the top sensor channels, at 50 eV amplifies the already severe task difficulty for the OF-transformer leading to the largest increase in σ/E_R .

At $E_R = 10$ eV the increase in σ/E_R when masking the top and bottom channels is comparable (see Figure 12, Table 5). The similarity in the degradation at this recoil energy occurs despite the entirety of the signal information lying in the top sensor channels at 10 eV. We attribute this to the OF-transformer model’s cross channel attention [27] emphasising the global magnitude of the sensor channels’ amplitude-like representations. Without the noise floor baseline that the model is trained on in the bottom sensor channels the model predicts E_R poorly. The effect is most pronounced at 10 eV as the signal to noise ratio is lower and therefore the relative degradation when masking bottom sensor channels is higher. However, this requires further investigation where instead of setting masked values to zero masked inputs are replaced with noise only traces.

We observe at $E_R = 10$ eV the OF-transformer achieving a lower RMSE_{pos} with bottom masking than with no masking at all. At $E_R = 10$ eV the bottom sensor channels contain entirely noise and so by masking them out we assert an implicit inductive bias instructing the transformer to focus only on the relational information between top sensor channels, where all the information is contained. At higher energies information starts to be distributed between top and bottom sensor channels (See Figure 3) [7] and so bottom sensor channel masking becomes disadvantageous for position reconstruction. This is evidenced by the unmasked configuration achieving a lower RMSE_{pos} than the other two masking configurations for $E_R \geq 20$ eV.

The impact of top sensor channel masking on position reconstruction decreases with increasing E_R . This aligns with the transition in signal information away from the top sensor channels to between both top and bottom sensor channels. However, we observe a negligible effect on RMSE_{pos} when masking bottom sensor channels for all E_R . This suggests that it is the quasiparticle distribution, which is only recorded by top sensor channels [10], that is predominantly informative for position reconstruction.

ER/NR classification performance aligns with a uniform random classifier for all three masking configurations at $E_R = 10$ eV. Without access to meaningful signal channel ratios through excitations and ionisation irrespective of whether amplitude information is masked out or not the OF-transformer is unable to discriminate between ERs and NRs.

Notably ER/NR classification is the sole task where bottom sensor channel masking has a comparable impact on performance to top channel masking. Whilst for energy and position reconstruction we observe the largest degradation in performance with the largest fractional decreases in amplitude information adjusted for the difficulty of the E_R , for ER/NR classification it is the fraction of each signal type which serves to predominantly inform recoil type discrimination [7]. Consequently, despite in the aggregate most of the amplitude information lying in the top sensor channels, if the bottom sensor channels are masked when no scintillation is present in the top sensor channels then the available information is insufficient to inform ER/NR classification hence the impact on classification performance when masking bottom sensor channels.

The impact of sensor channel masking is both task and E_R dependent. Whilst top sensor channels inputs are critical for all three reconstruction tasks the extent of the degradation is observed to be most severe for position reconstruction and for energy reconstruction E_R with high signal partitioning variance. Conversely, bottom sensor channel masking has negligible impact on position reconstruction performance yet has comparable effects to top sensor channel masking on ER/NR classification at higher energies. The OF-transformer through the channel masking analysis has demonstrated that it has learned the underlying signal partitioning physics of the simulation outlined in [7].

5. Conclusion

The primary focus of this report has been on developing and evaluating the event reconstruction performance of transformer models against an OF-BDT baseline. Motivated by the geometry of the detector and the structure of the data we have successfully developed and trained transformer variants with and without OF amplitudes [22, 23] to conduct all three event reconstruction tasks.

We benchmarked the transformer variants against the OF-BDT baseline. By combining classical OF with modern deep learning methods, our OF-transformer outperforms the OF-BDT across all our primary reconstruction metrics across all recoil energies on the test dataset.

The OF-transformer achieves a resolution of $\sigma/E_R = 17.1\%$ against the OF-BDT baseline result of $\sigma/E_R = 20.8\%$ at $E_R = 20$ eV. This improvement over the OF-BDT will reduce the risk presented to DELight of misclassifying events at the Phase I detection threshold [11].

For position reconstruction the improvements in OF-transformer performance over the baseline are most stark. Whilst at 10 eV the OF-BDT is unable to accurately reconstruct event position with

$\text{RMSE}_{\text{pos}} = 33.9$ mm, the OF-transformer achieves an RMSE_{pos} of 5.85 mm. The OF-transformer will therefore enable characterisation of events at and below the current Phase I threshold [11] and by extension permit rejection of otherwise irreducible background, potentially extending the sensitivity limits of the experiment.

We observe the smallest impact on the broader experiment for ER/NR classification. Whilst the OF-transformer achieves a larger overall ROC-AUC of 0.945 against that of the OF-BDT's 0.893, the classification performance at energies $E_R \leq 20$ eV remains poor across all models due to constraints imposed by the signal partitioning physics. Consequently, it is for energies $E_R \geq 50$ eV where both transformer variants will facilitate more reliable background rejection for DELight.

Having systematically evaluated the performance of the OF-transformer on the test set for top masking, bottom masking and unmasked sensor configurations, we demonstrate the model's capacity to learn the simulation's underlying signal partitioning physics [7]. This provides the DELight experiment [11] with a framework for interpreting the behaviour of the transformer variants. We have also highlighted through channel masking the overwhelming dominance of quasiparticle signatures in position reconstruction.

Future work on channel masking analysis is required to identify the mechanisms driving comparable resolution performance degradation between top and bottom sensor channel masking at 10 eV. Additionally, whilst the training data is injected with experimentally derived noise PSDs [23], model evaluation on experimentally obtained traces is required to validate the results of this report.

In having adapted transformers [27] to better perform event reconstruction than the OF-BDT baseline we potentially facilitate DELight's ability to exclude sub-GeV regions of the DM direct detection parameter space. The architectural construction presented here may additionally prove adaptable to any LDM detector geometry seeking to leverage trace data for reconstruction. The extent to which this holds true would serve as a potential subject of future investigation.

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